

Self-organized epigenetic signal processing in active chromatin

Active histone marks turn over rapidly, yet chromatin can retain memory of prior stimulation. We develop a theoretical and microscopic polymer framework showing how state-dependent spatial amplification of regulatory factors generates nonlinear feedback and two thresholds separating OFF, MEMORY, and ON regimes. In simulations, marked chromatin and mobile factors self-organize into localized reaction hubs, producing the required nonlinear amplification. We develop a theory that captures the state-dependent amplification measured from simulations and use it to predict distinct OFF, MEMORY, and ON regimes, as well as recovery dynamics, first-passage statistics, and frequency-dependent hysteresis. Strong signals establish marks, weaker signals maintain them, and repeated signals can accumulate over time. Our results suggest that self-organized spatial feedback provides a simple physical mechanism for epigenetic memory and signal processing in active chromatin.